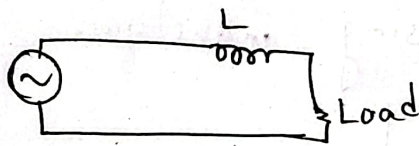


① Is it possible to make Low pass filters using an inductor?

⇒ So first we need to understand how we can fix the inductor to the circuit. Is it series connection or parallel connection? Let us understand how an inductor works.



Series

In series with AC current, the current always fluctuates. Means, the circuit will be magnetic when it gets high frequency. When it gets low freq, it acts as short circuit.

mean,

Low frequency $\Rightarrow$ pass

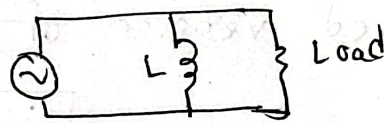
Act as short circuit

High frequency = blocks the

AC, pass the DC. So

we can understand that,

A series inductor can be used to make filters to block high freq and pass Low freq.



In parallel connection, If Low frequency $\Rightarrow$ Acts as short circuit and will be dangerous because AC source is directly connected.

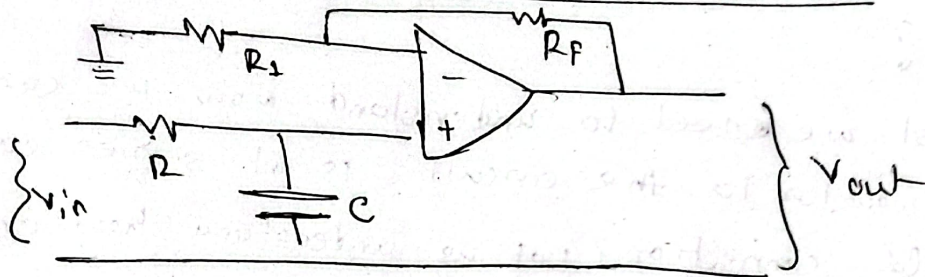
High frequency of inductor Acts as a magnet, needs more emf to break the barriers of X_L . Then it will use full current and block the load.

So, best way is to choose Series inductor to make filters.

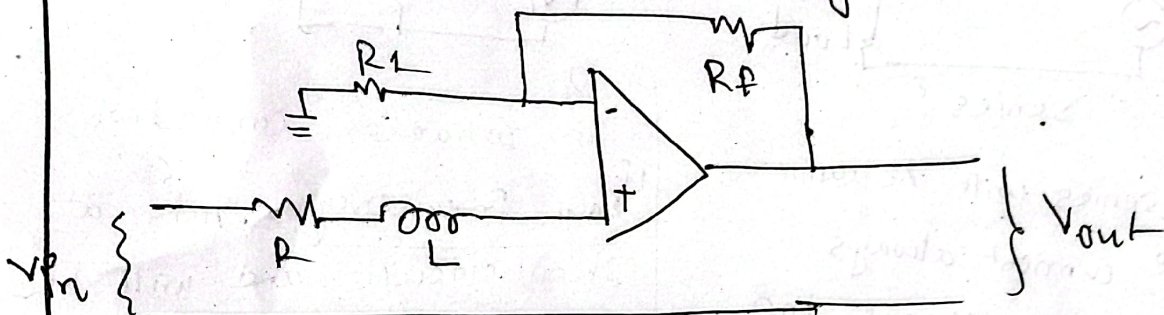
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A Low pass filter using capacitor



modified version of this using inductor



see that, the inductor is series connected.

Mathematically:

The impedance of inductor, $Z_L = j\omega L \Rightarrow j2\pi f L$

The transfer function,
$$-\frac{V_{out}}{V_{in}} = \frac{R}{R + j\omega L}$$

and the magnitude,

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{R}{\sqrt{R^2 + (\omega L)^2}}$$

so now we analyse this according to low frequency and high frequency.

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At Low frequency, $f \Rightarrow 0$
or, $\omega = 0$

$$\text{So, } \left| \frac{v_{out}}{v_{in}} \right| \Rightarrow \frac{R}{\sqrt{R^2 + 0}} \Rightarrow 1$$

means Low frequency will pass.

At high frequency, $f = \infty$
 $\omega = \infty$

$$\left| \frac{v_{out}}{v_{in}} \right| \Rightarrow \frac{R}{\sqrt{\omega^2 L^2}} = 0$$

so, high freq will be blocked.

But practically :

$$\text{cutoff frequency, } f_c = \frac{R}{2\pi L}$$

$$\text{or, } L \Rightarrow \frac{R}{2\pi f_c}$$

Suppose,

$$f_c \Rightarrow 1 \text{ kHz}$$

$$R \Rightarrow 10 \text{ k}\Omega$$

$$L \Rightarrow \frac{10,000}{2 \times \pi \times 1000} = 1.6 \text{ H}$$

1.6H is a huge value, it is ~~impossible~~
impossible to generate this value.

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The 2nd problem we will face is voltage,

$$\text{we know, } V \Rightarrow L \frac{dI}{dt}$$

For AC current and high frequency, $\frac{dI}{dt}$ is high.

$$\text{So, } L \Rightarrow 1.6 \text{ H}$$

So voltage will be large value. But opamp can't handle this level of voltage.

So, mathematically, it is possible, but practically it is impossible.

But why capacitor is suitable for this?

$$\Rightarrow \text{we know, } f_c \Rightarrow \frac{1}{2\pi R_c C}$$

$$R \Rightarrow 10,000 \Omega$$

$$f_c \Rightarrow 1000 \text{ Hz}$$

$$C \Rightarrow \frac{1}{2\pi \times 10,000 \times 1000}$$

$$\Rightarrow 40 \text{ nF}$$

This capacitor is cheap, stable and IC-friendly. Capacitors can't generate noise but filter it. But inductor makes noise, attenuation at high frequency. That's why capacitor is perfect for low pass filter.

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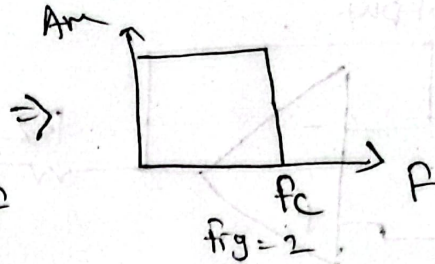
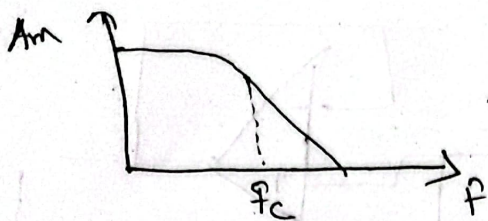
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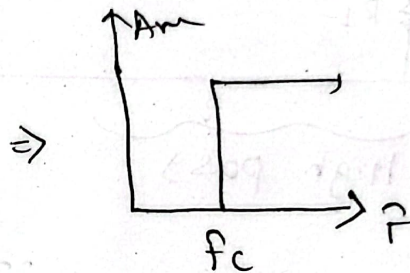
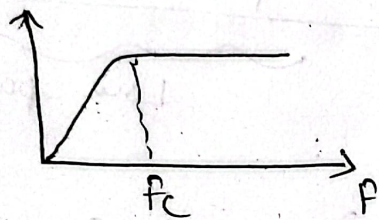
(2) Implementation of Band pass filter.

⇒ first we need to understand the Low pass and high pass curve.

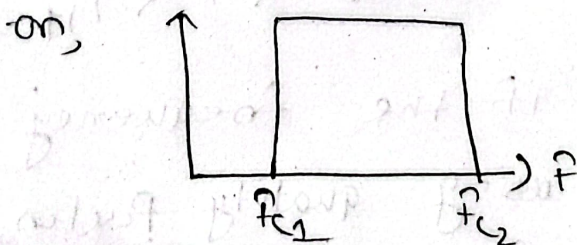
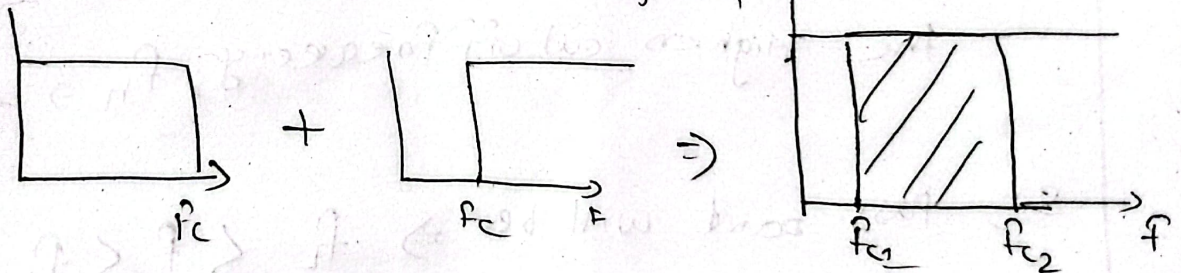
Low pass filter ⇒



High pass filter ⇒



So if we merge both diagram ⇒

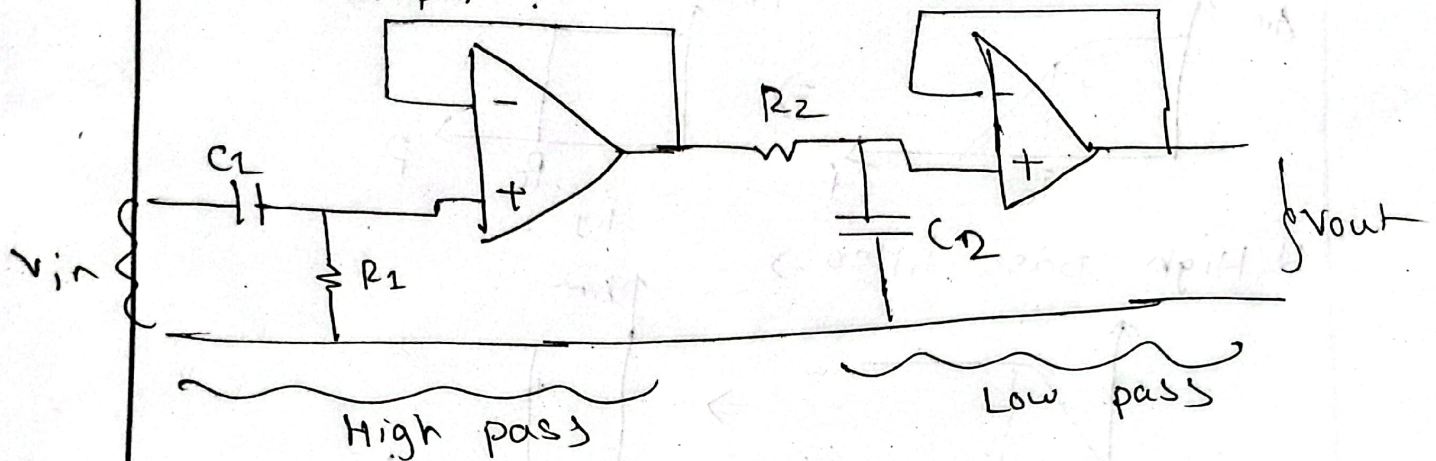


By this curve, Low and high frequency will be terminated but the middle frequency will be output.

So, the Bandwidth will be $\Rightarrow f_{c2} - f_{c1}$

center frequency $\Rightarrow \sqrt{f_{c1} \times f_{c2}} = f_0$

Center frequency means where ω_c will get the max output.



So, the lower cut-off frequency, $f_{cL} \Rightarrow \frac{1}{2\pi R_1 C_1}$

the higher cut-off frequency, $f_{cH} \Rightarrow \frac{1}{2\pi f_2 C_2}$

So, Pass band will be $\Rightarrow f_L < f < f_{Ht}$

So, we can check if the frequency is narrow or wide, by using quality factor, Q .

$Q \Rightarrow \frac{\text{Center frequency}}{\text{Bandwidth}}$

Q large $\Rightarrow$ narrow band

Q small $\Rightarrow$ wide band

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So, from this estimation, we can say,

If,

- $f < f_L$ then output 0
- $f = f_L$ then output multiply of 0.707
- $f = f_0$ then output max
- $f = f_H$ then output multiply of 0.707
- $f > f_H$ then output 0

then the phase behavior will be.

If,

- $f < f_L$ then output 90°
- $f = f_0$ then " 0°
- $f > f_H$ " " -90°

So, the curve will be

